CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

Third Semester of B. Tech. (CL) Examination
November 2010

CL 205 Building Materials and Construction

Date: 01.12.2010, Wednesday **Time: 10:30 a.m. To 01:30 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Listout properties of engineering materials? Explain any four in detail. [5]

(b) Enlist types of glass. [2]

Q - 2 (a) Which building materials are used as finishing or covering materials? Write their specific uses. [6]

(b) What are the properties of fresh and hardened concrete? [5]

(c) Write factors affecting selection of flooring material. [3]

OR

Q - 2 (a) Write the composition of cement. Differentiate between quick setting cement and rapid hardening cement. [6]

(b) Write types and uses of industrial timber. [5]

(c) Draw a neat sketch of timber flooring. [3]

Q - 3 (a) Design a dog-legged stair for 3.0 m floor height. The hall size is 2.5 m x 5.5 m. Assume suitable data and states it clearly. Prepare sketch with dimensions. [7]

(b) What are the types of roof and roof covering? What are the merits and demerits of flat roof ? [5]

(c) Enlist different types of flooring. [2]

OR

Q - 3 (a) Sketch various types of steps. What are the requirements of stairs? [7]

(b) Draw neat sketch of king post truss with appropriate labeling. [5]

(c) Distinguish between ground floor and upper floor. [2]

SECTION – II

Q - 4 (a) Write short notes on (i) Defects in painting (ii) Random rubble un-coursed masonry [5]

(b) What are the requirements of a good plaster? [2]

Q - 5 (a) Draw elevation and plan of alternate courses of English bond for 1½ brick thick wall. [5]

(b) Differentiate between Acute squint quoin and Obtuse squint quoin at connections in brick masonry. [2]

(c) Design a strip footing for a brick wall 30 cm thick and 3.4 m high above the ground level. The wall carries a superimposed load of 115 kN/m run. The soil has unit weight of 17 kN/m^3, angle of repose of 33^o and safe bearing capacity of 171 kN/m^2. The footing may have lime concrete base, which has unit weight of 20 kN/m^3 and modulus of rupture equal to 154 kN/m^2. Assume, unit weight of masonry is 19.5kN/m^3. [7]

OR

Q - 5 (a) Draw elevation and plan of alternate courses of Double Flemish bond for 1 brick thick wall. [5]

(b) Write short note on stone facing with brick backing masonry. [2]

(c) What are the causes of failures of foundation and suggest remedial measures? [3]

(d) Write advantages and disadvantages of pre-cast concrete piles. [4]

Q - 6 (a) Discuss the points to be considered while locating doors and windows. Draw a neat sketch of solid core flush door. [6]

(b) Why ventilator is provided? Give the standard sizes of windows and ventilators. [3]

(c) Write short note: (Any Two) [3]

(i) Vertical sheeting (ii) Single flying shore (iii) Needle scaffolding

(d) What are the essential requirements of a good foundation? [2]

OR

Q - 6 (a) Define: Mullion, Transom and Horn. Draw neat sketch of bay window. [6]

(b) Write short note on fan light ventilator. [3]

(c) Write short note: (Any Two) [3]

(i) Runner system (ii) Mason's scaffolding (iii) Pile method of underpinning

(d) Write short note on simplex standard pile. [2]

****